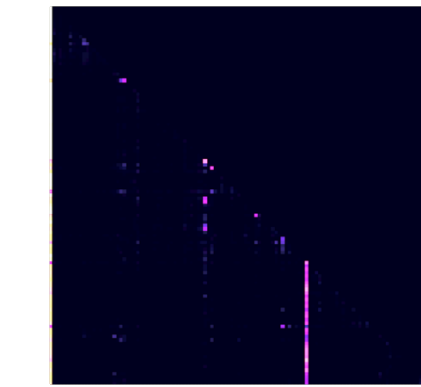
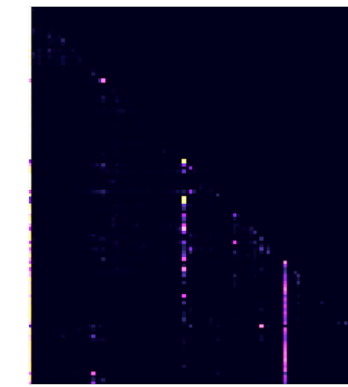
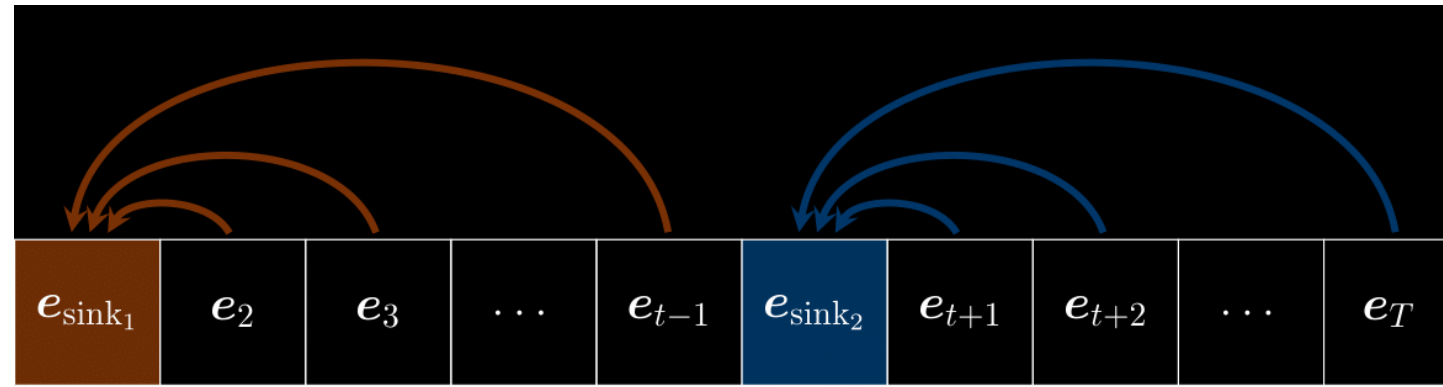
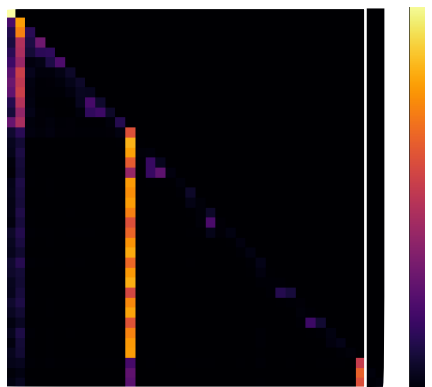
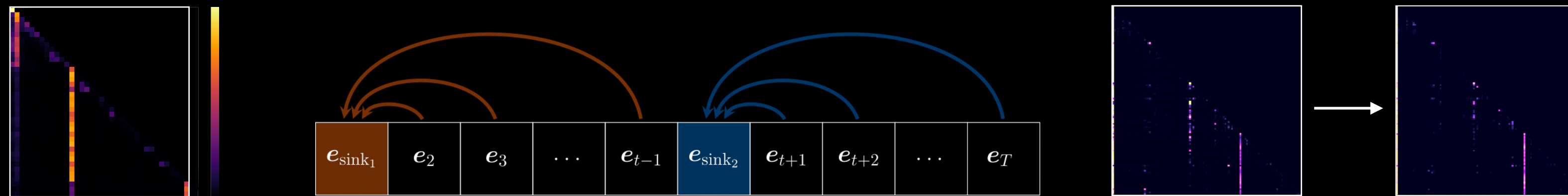




Summary

- Low rank structure in W_Q , W_K , W_V matrices



- Emergence of attention sinks + outlier features
- Theoretical framework for catch, tag, release
- Concept-based sentiment analysis
- Exploit for outlier-aware pruning
 - Outperforms existing methods

Implications + Future Work

- Early exit and stochastic depth
- Layer pruning and merging
- Model compression through weight projection
- Explains invariance under block permutations
- Model merging through alignment of coupled subspaces
- Promoting coupling during training through Unconstrained Jacobians
- Promoting low rank structure during training
- Exploration of significance of smallest singular values